AAA-S-118D
April 30, 1974
SUPERSEDING
Fed. Spec. AAA-S-118C
September 29, 1970

FEDERAL SPECIFICATION

SCALES, BEAM INDICATING (PORTABLE-PLATFORM, FLOOR TYPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers floor type, portable platform scales.

1.2 Classification. The scales covered by this specification shall be of the following types, styles, and capacities as specified (see 6.2).

Type I - Conventional type.

Style 1 - Single beam.

Capacities:

500 pound
1000 pound
1500 pound
2000 pound
2500 pound

Type III - Folding type.

Style 1 - Single beam.

Capacities:

600 pound
1200 pound

Style 3 - Full capacity beam.

Capacity:

300 pound

FSC 6670

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-1122 - Balances, Scales, and Accessories, Packaging and Packing of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Bureau of Standards (NBS) Standard

Handbook 44 - Specification, Tolerances, and Other Technical Requirements for Commercial Weighing and Measuring Devices

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the scale to be procured under this specification for first article inspection and approval (see 4.2 and 6.3).

3.2 Standard product. The scale, shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modification to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the government consistent with their normal commercial practices.

3.3 Materials. Material not definitely specified shall be of the quality normally used by the manufacturer for scales, provided the completed item complies with all the provisions of this specification.

3.4 Design and construction. The type I scale shall be similar to figure 1. The type III scale shall be similar to figure 3. Figures are intended for illustration purposes only, and are not intended to preclude scales otherwise meeting the requirements of this specification. Scales and counterpoise weights shall conform to the applicable general and user requirements of National Bureau of Standards Handbook 44.

3.5 Details.

3.5.1 Type I scale. The type I scale shall be a conventional, portable, beam scale. The scale shall consist of a weighbeam mounted on a metal pillar attached to a frame. The scale shall be provided with four metal wheels. When specified (see 6.2), a device shall be provided to prevent platform forces being transmitted to the lever system. Counterpoise weights shall be supplied as specified in table I.

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TABLE I - Counterpoise weights, type I, style 1 scales

Capacity in pounds	Counterpoise weights Quantity - Size
500	1 - 200
	2 - 100
	1 - 50 1/
1000	1 - 500
	1 - 200
	2 - 100
	1 - 50 1/
1500	2 - 500
	1 - 200
	2 - 100
2000	3 - 500
	1 - 200
	2 - 100
2500	4 - 500
	1 - 200
	2 - 100

1/ Required when 50 lb. weighbeam is furnished.

3.5.1.1 Platform, type I. Unless otherwise specified (see 6.2), the platform of the type I scale shall have either a cast iron, steel or aluminum frame, with a wood, steel, aluminum or cast iron cover plate, at the option of the manufacturer. Platform dimensions shall conform to table II.

TABLE II - Type I conventional scales

Scale capacity	Platform dimensions minimum, inches		Weighbeam capacity and subdivisions	
			Capacity graduation	
Pounds	Rectangular platform	Square platform 1/	Pounds	Pounds
500	16 x 24	24 x 24	50 or 100	1/4 or 1/2, respectively
1000	17 x 26	26 x 26	50 or 100	1/4 or 1/2, respectively
1500	21 x 30	30 x 30	100	1/2
2000	23 x 30	28 x 28	100	1/2
2500	25 x 34	28 x 28	100	1/2

1/ When a square platform is required (see 6.2).

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3.5.1.2 Weighbeams, type I. The weighbeams shall have a capacity and graduations specified in table II. The weighbeam for the style 1 scale shall be a single, smooth bar, graduated on both sides.

3.5.2 Type III scale. The type III scale shall be the platform folding type. The scale shall be designed and constructed so that, without disassembly, the upright weighbeam assembly shall fold down on the platform to allow the scale box to enclose the entire scale mechanism.

3.5.2.1 Platform, type III. The platform for the type III scale shall be constructed either with a cast iron frame or steel frame. The platform shall have a steel center. Platform dimensions shall be as specified in table III.

TABLE III - Type III platform dimensions

Scale capacity	Platform dimensions minimum
Pounds	Inches
300	13 x 17
600	17 x 22
1200	19 x 28

3.5.2.2 Scale box. The scale box shall be constructed either of hard wood, reinforced at the corners with metal, covered on the top and sides of the upper portions with sheet metal, or shall be of all metal construction. The lower section of the scale box, enclosing the lever system and platform, shall be hinged to the upper section of the scale box, enclosing a beam rod and weighbeam assembly. A safety hasp and staple shall be provided for locking the box in a closed position. A valance shall be secured to the upper section of the box. The valance shall project not less than 1-inch, to cover the joint on the sides and the unhinged end when the box is closed. A hinged door shall be provided at the top of the upper section of the box to allow the weighbeam to be operated and read from the back of the box when the scale is set up. A door latch shall be provided, operable only from the inside of the box. Two chest handles shall be provided on each side of the lower section of the box. Skid cleats shall be provided on the top and bottom of the box. The skid cleats shall be set parallel and, either flush with the long-edges, or set in from the edges, at the option of the manufacturer. A stop with a lock shall be provided for securing the upper section of the box in a vertical position.

3.5.2.3 Securing of components. The scale shall be designed and assembled to prevent any damage or derangement of components when the scale is transported with the box closed when tested as specified in 4.4.6. When the scale is opened into weighing position and the safety latches are released, the scale will be in operating condition without any special adjustments (see 4.4.6).

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Safety latches shall be provided to secure the weighbeam and the counterpoise hanger supplied with the style 1 scale. Safety latches shall also be provided to support and raise the platform from its bearings and to secure the platform when scale is moved. The platform bearings shall be designed to prevent the platform from slipping out of the bearings when the scale is tilted or tipped over. Means shall be provided for securing counterpoise weights for style 1 scales.

3.5.2.4 Weighbeam, type III.

3.5.2.4.1 Weighbeams, type III, style 1 scales. The weighbeams for type III, style 1 scales shall consist of a single, smooth bar with graduations on both sides. The poise shall be equipped with a locking screw. The weighbeam for the 600 pound capacity scale shall be graduated to 50 pounds by 1/4 pound subdivisions. The weighbeam for the 1200 pound capacity scale shall be graduated to 100 pounds by 1/2 pound subdivisions. Counterpoise weights shall be supplied as specified in table IV.

TABLE IV - Counterpoise weights, type III, style 1 scales

Capacity in pounds	Counterpoise weights Quantity - Size
600	2 - 200
	1 - 100
	1 - 50
1200	1 - 500
	2 - 200
	2 - 100

3.5.2.4.2 Weighbeams, type III, style 3. scales. The weighbeam for the type III, style 3 scale shall be a full capacity weighbeam consisting of a main bar and a fractional bar. Both sides of the bars shall be graduated. The main bar shall be notched and the fractional bar shall be smooth. The main bar for the 300 pound capacity scale shall be graduated to 275 pounds by 25 pound subdivisions. The fractional bar shall be graduated to 25 pounds by 1/4 pound subdivisions. Poises on all bars shall be equipped with locking screws.

3.6 Performance.

3.6.1 Capacity. When a load equal to 110 percent capacity of the scale is placed on the platform, the scale shall show no indication of permanent deformation, bends, breakage, separation, or misalignment of parts, and shall return to the initial zero position after the load is removed, when tested as specified in 4.4.1.

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3.6.2 Sensitivity requirement. The sensitivity requirement shall be in accordance with H-44 for scales to which acceptance tolerances shall apply, when tested as specified in 4.4.5.

3.6.3 Acceptance tolerances (scales). The acceptance tolerances shall be in accordance with H-44 for large capacity scales when tested as specified in 4.4.2 and 4.4.4.

3.6.4 Acceptance tolerances (counterpoise weights). The acceptance tolerances shall be in accordance with H-44 for counterpoise weights when tested as specified in 4.4.3.

3.7 Finish. The scales shall have all surfaces finished in accordance with the manufacturer's current commercial practice.

3.8 Caution notice for type III. A decal shall be attached to the upper section of the box for the type III scale containing the following:

"CAUTION

TO PREVENT DAMAGE,

SECURE THE SAFETY LATCHES ON THE WEIGHBEAM

SYSTEM AND ON THE PLATFORM BEFORE

MOVING THIS SCALE"

3.9 Marking for identification. The manufacturer's name, trade name, or trademark of such known character as to be identifiable, and the model number shall appear on each scale. The standard commercial product provided under this specification shall be identifiable by the manufacturer's regular or commercial parts distribution or service organizations.

3.10 Repair and maintenance. Provisions shall be made to allow for adjustment, service, and replacement of components. In addition, there shall be access to inspect, service, and adjust any component as specified herein or as indicated by the component or equipment manufacturer. Replacement or repair parts, and service for the standard commercial product provided under this specification shall be available from the manufacturer's regular or through commercial parts distribution or service organizations.

3.11 Workmanship. The finished scales shall have no burrs, slivers, splinters, or broken, loose, or missing fasteners.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, (see 6.2), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When a first article is required, it shall be examined for the defects in table V, specified dimensions, and tested in accordance with 4.4.1 through 4.4.5, as applicable. The type III scales shall be tested in accordance with 4.4.6, prior to other tests. The presence of any visual defect, any dimension not within the specified requirements, or failure to pass any of the tests shall be cause for rejection of the first article.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 In-process inspection. The scales and weights shall be inspected for compliance with the applicable requirements of H-44 (see 3.4). Whenever a deviation is noted, correction shall be made to the affected items and lot in process.

4.3.2 End item inspection. A lot shall consist of all the scales of one type, style, and capacity offered for inspection at one time. The sample unit shall be one completely fabricated scale.

4.3.2.1 Visual examination. The end item shall be examined for the defects classified in table V. The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE V - Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's current commercial finish.		X
Construction and workmanship (general)	Not type, style or capacity specified.	X	
	Not in accordance with the general and user (except dimensional) requirements of H-44.	X	
	Any component not readily accessible for servicing, when required.	X	
	Burr, sliver, or splinter.		X
	Broken, loose, or missing fastener.		X
Platform	Not as specified.	X	
Type I	Not provided with wheels.	X	
	Counterpoise weights not furnished, as specified.		X
	Weighbeam not graduated, as specified.	X	
Type III	Scale box door not as specified.	X	
	Valance or skid cleats missing.	X	
	Scale box does not inclose entire scale mechanism.	X	
	Two chest handles not furnished on box.		X
	Weighbeam not graduated as specified.	X	
	Poise not provided, or locking screw missing.	X	
	Safety latches not as specified.	X	
	Counterpoise weights not furnished or secured, when required.	X	
	"Caution" decal not correct or missing.		X
Marking for identification	Missing, incomplete, not legible.		X

4.3.2.2 Dimensional examination. Examination shall be made of the end item for compliance with dimensions specified and as specified in H-44. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

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4.3.2.3 Testing of the end item. The end item shall be tested as specified in 4.4.1 through 4.4.5, as applicable. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units. When a first article of type III scales is not required, the initial unit of production shall be tested as specified in 4.4.6, prior to other testing. Failure of this test shall be cause for rejection of the lot.

4.3.3 Examination of preparation for delivery. Examination of preparation for delivery shall be in accordance with PPP-B-1122.

4.4 Tests.

4.4.1 Overload test. Weights equal to 110 percent of capacity shall be placed on the platform of the scale and then removed. Any noncompliance with the requirements of 3.6.1 shall constitute failure of this test.

4.4.2 Shift test. Loads equal to 1/4 capacity, shall be successively placed over each main load support of the platform. Any noncompliance with the requirements of 3.6.3 shall constitute failure of this test.

4.4.3 Counterpoise weight test. The counterpoise weights shall be weighed. Any noncompliance with the requirements of 3.6.4 shall constitute failure of this test.

4.4.4 Increasing load test. The tolerance of the scale at increasing loads shall be determined at 1/8, 1/4, 1/2, and full capacity of the scale. Place a weight equal to each of these loads on the center of the platform successively. Any noncompliance with the requirements of 3.6.3 shall constitute failure of this test.

4.4.5 Sensitivity test. The test for sensitiveness shall be conducted on all nonautomatic-indicating scales. SR tests shall be made at zero load and at the maximum test load applied to the scale be either increasing or decreasing by a specified amount (see SR. Sensitivity Requirements) the test-weight load on the load-receiving element of the scale. The minimum response of the scale shall be as follows: The position of rest of the weighbeam shall change from the center of the trig loop to the top or bottom, as the case may be. Any noncompliance with requirements of 3.6.2 shall constitute failure of this test.

4.4.6 Road test (type III). Prior to the tests specified in 4.4.1 through 4.4.5, the closed type III scale shall be loaded into a truck. The truck with the scale shall be driven a distance of 15 miles consisting of not less than 5 miles on a paved road at speeds up to 50 miles per hour (mph); over not less than 5 miles of graveled roads at speeds averaging 35 mph; and over not less than 5 miles of open terrain and cross country at speeds averaging 5 mph. The scale shall then be removed from the truck and examined. Any noncompliance with the requirements of 3.5.2.3 shall constitute failure of this test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-B-1122 (see 6.2). Preservation and packaging shall be level A or C, and packing shall be level A, B or C (see 6.2).

6. NOTES

6.1 Intended use. The type I scale covered by this specification is intended for general purpose weighing. The type III scale is intended for use in the field by the military.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, style, and capacity required (see 1.2).
(c) When a first article is required (see 3.1, 4.2, and 6.3).
(d) When a device is required to prevent platform forces being transmitted to the lever system (see 3.5.1).
(e) When a specified platform construction is required (type I (see 3.5.1.1).
(f) When a square platform is required (type I) (see table I).
(g) Level of preservation and packaging, and level of packing required (see 5.1).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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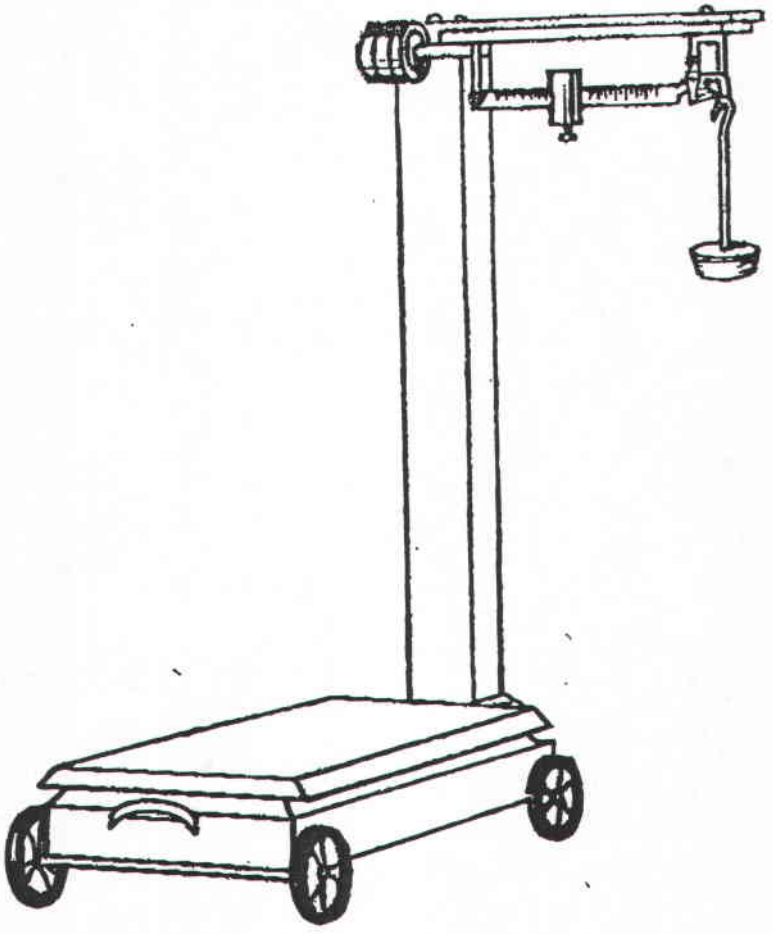


FIG. 1 - TYPE I, CONVENTIONAL TYPE, PORTABLE PLATFORM SCALE

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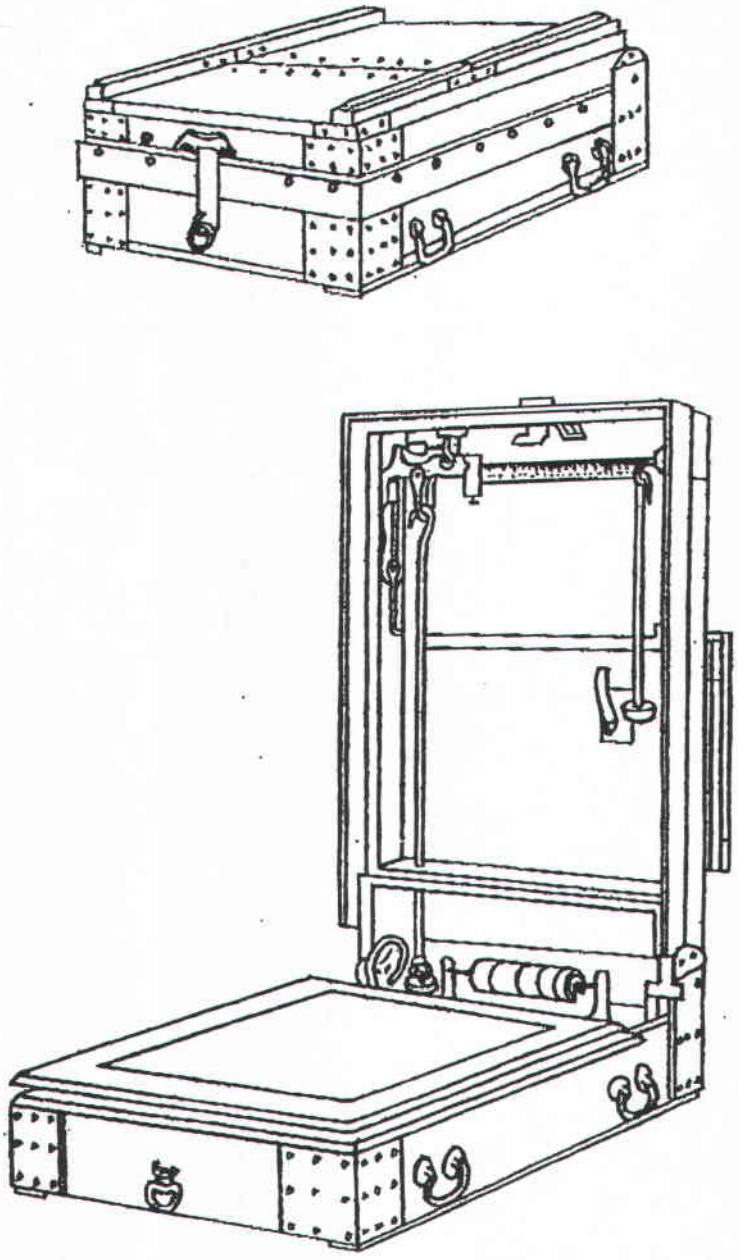


FIG. 3 - TYPE III, FOLDING TYPE, PORTABLE PLATFORM SCALE

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Custodians:

Army - GL
Navy - YD
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Review activities:

Navy - MC

Preparing activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

COM-NBS
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